

MATH3904 Introduction to Optimization**§1. Basic Properties of Solutions and Algorithms**

Consider the following optimization problem

$$\begin{array}{ll} \text{Minimize} & f(x_1, x_2, \dots, x_n) \\ \text{subject to} & (x_1, x_2, \dots, x_n) \in \Omega, \end{array}$$

where f is a real-valued function, Ω is a subset of $\mathbb{R}^n$, and $\mathbb{R}^n$ is the n -dimensional space of all real vectors.

In vector notation, the above problem becomes

$$\begin{array}{ll} \text{Min} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in \Omega. \end{array} \quad (1.1)$$

Usually, f is called the objective function, Ω is called the feasible region, and every $\mathbf{x}$ in Ω is called a feasible point (or feasible solution). A point $\mathbf{x}^*$ in Ω is called an optimal solution to (1.1) if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for any $\mathbf{x} \in \Omega$, and $f(\mathbf{x}^*)$ is called the optimal value of (1.1).

The problem (1.1) is called an unconstrained optimization problem if $\Omega = \mathbb{R}^n$ and constrained optimization problem otherwise.

The purpose of this course is to develop efficient algorithms for solving various problems of the form (1.1). Now let us present some well-solved examples.

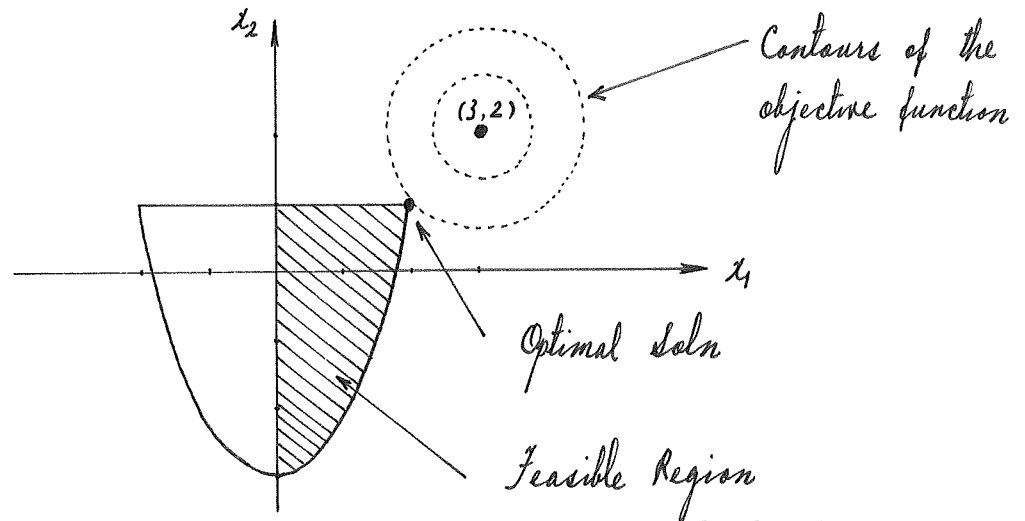
EXAMPLE 1.1. Geometric Method

$$\begin{array}{ll} \text{Min} & (x_1 - 3)^2 + (x_2 - 2)^2 \\ \text{s.t.} & x_1^2 - x_2 - 3 \leq 0 \\ & x_2 - 1 \leq 0 \\ & x_1 \geq 0. \end{array}$$

SOLUTION. The problem is to find a point in the feasible region with the smallest possible value of $(x_1 - 3)^2 + (x_2 - 2)^2$.

Let c stand for $(x_1 - 3)^2 + (x_2 - 2)^2$. Then $(x_1 - 3)^2 + (x_2 - 2)^2 = c$ represents a circle with radius $\sqrt{c}$ and center $(3, 2)$. The circle is called the contour of the objective function having the value c . Since we wish to minimize c , we must find the circle with the smallest radius that intersects the feasible region. As shown in the following figure, the smallest such circle has $c = 2$ and intersects the feasible region at the point $(2, 1)$. Therefore the

optimal solution occurs at the point (2, 1) and has an objective value equal to 2. $\square$



The approach used above is to find an optimal solution by determining the objective contour with the smallest objective value that intersects the feasible region. Obviously, this approach of solving the problem geometrically is only suitable for small problems and is not practical for problems with more than three variables or those with complicated objective and constraint functions.

EXAMPLE 1.2. Algebraic Method

Let $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix. Solve the problem

$$\begin{aligned} \text{Min } & \mathbf{x}^T A \mathbf{x} \\ \text{s.t. } & \mathbf{x}^T \mathbf{x} = 1. \end{aligned}$$

SOLUTION. Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ be all the eigenvalues of A . Since A is symmetric, by a theorem in linear algebra, A can be diagonalized by using an orthogonal matrix Q , that is

$$A = Q^T \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{pmatrix} Q.$$

Let $\mathbf{y} = Q\mathbf{x}$. Then $\mathbf{y}^T \mathbf{y} = \mathbf{x}^T Q^T Q \mathbf{x} = \mathbf{x}^T \mathbf{x} = 1$. Thus

$$\begin{aligned} \mathbf{x}^T A \mathbf{x} &= \mathbf{x}^T Q^T \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} Q \mathbf{x} = \mathbf{y}^T \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \mathbf{y} \\ &= \sum_{i=1}^n \lambda_i y_i^2 \\ &\stackrel{(1)}{\geq} \lambda_n \sum_{i=1}^n y_i^2 = \lambda_n \mathbf{y}^T \mathbf{y} = \lambda_n. \end{aligned}$$

Note that (1) holds equality if $\mathbf{y} = (0, 0, \dots, 0, 1)^T$. Hence the optimal value is λ_n which is achieved when $\mathbf{y} = (0, 0, \dots, 0, 1)^T$ or $\mathbf{x} = Q^T \mathbf{y} = Q^T \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$. $\square$

QUESTION 1.1. What is the optimal value of the corresponding maximization problem?

QUESTION 1.2. Suppose A is an arbitrary square matrix. What is the solution?

EXAMPLE 1.3. Solve the following problem

$$\begin{array}{ll} \text{Min} & f(x_1, x_2) = x_1^2 - x_1 + x_2 + x_1 x_2 \\ \text{s.t.} & x_1 \geq 0, \quad x_2 \geq 0. \end{array}$$

SOLUTION. Clearly $f(x_1, x_2) = (x_1 - \frac{1}{2})^2 + x_2(1 + x_1) - \frac{1}{4}$. Since for all feasible points (x_1, x_2) , we have $(x_1 - \frac{1}{2})^2 \geq 0$ and $x_2(1 + x_1) \geq 0$. Hence $f(x_1, x_2) \geq -\frac{1}{4}$. Note that setting

$$\begin{cases} (x_1 - \frac{1}{2})^2 = 0 \\ x_2(1 + x_1) = 0 \end{cases}$$

yields

$$\begin{cases} x_1 = \frac{1}{2} \\ x_2 = 0. \end{cases}$$

Hence the optimal value is $-\frac{1}{4}$, which occurs on $(\frac{1}{2}, 0)$. $\square$

EXAMPLE 1.4. Find an optimal solution to the problem

$$\begin{array}{ll} \text{Min} & f(\mathbf{x}) = \sum_{i=1}^n \frac{c_i}{x_i} \\ \text{s.t.} & \sum_{i=1}^n a_i x_i = b \\ & x_i \geq 0 \quad \text{for } i = 1, 2, \dots, n, \end{array}$$

where a_i, b, c_i are all positive constants for $i = 1, 2, \dots, n$.

SOLUTION. Our approach relies on the famous Schwartz inequality: Let $\mathbf{g} = (g_1, g_2, \dots, g_n)^T$ and $\mathbf{h} = (h_1, h_2, \dots, h_n)^T$ be two nonzero vectors in $\mathbb{R}^{n \times n}$. Then $(\mathbf{g}^T \mathbf{h})^2 \leq \|\mathbf{g}\|^2 \|\mathbf{h}\|^2$ and equality holds iff $\mathbf{g}$ is a constant multiple of $\mathbf{h}$.

Clearly $f(\mathbf{x}) = \frac{1}{b} (\sum_{i=1}^n a_i x_i) (\sum_{i=1}^n \frac{c_i}{x_i}) = \frac{1}{b} \|\mathbf{g}\|^2 \|\mathbf{h}\|^2$, where $\mathbf{g} = (\sqrt{a_1 x_1}, \dots, \sqrt{a_n x_n})^T$ and $\mathbf{h} = (\sqrt{\frac{c_1}{x_1}}, \dots, \sqrt{\frac{c_n}{x_n}})^T$. By the Schwartz inequality, $\|\mathbf{g}\|^2 \|\mathbf{h}\|^2 \geq (\mathbf{g}^T \mathbf{h})^2 = (\sum_{i=1}^n \sqrt{a_i c_i})^2$ and with equality iff $\mathbf{g} = k\mathbf{h}$ for some constant k . It follows that $f(\mathbf{x}) \geq \frac{1}{b} (\sum_{i=1}^n \sqrt{a_i c_i})^2$ and equality holds iff $\sqrt{a_i x_i} = k \sqrt{\frac{c_i}{x_i}}$ or $x_i = k \sqrt{\frac{c_i}{a_i}}$ since $x_i \geq 0$ for $i = 1, 2, \dots, n$. Now

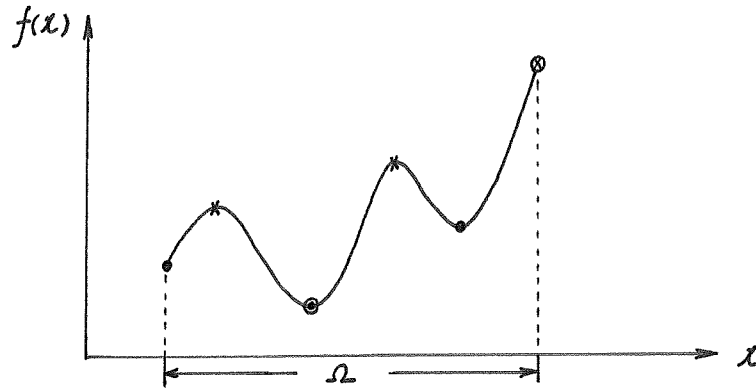
plug x_i into the equality constraint, we have $k \sum_{i=1}^n \sqrt{a_i c_i} = b$ and thus $k = \frac{b}{\sum_{i=1}^n \sqrt{a_i c_i}}$. From

the above arguments, we can conclude that the optimal value $\frac{1}{b} (\sum_{i=1}^n \sqrt{a_i c_i})^2$ is achieved at $\mathbf{x} = k(\sqrt{\frac{c_1}{a_1}}, \dots, \sqrt{\frac{c_n}{a_n}})^T = \frac{b}{\sum_{i=1}^n \sqrt{a_i c_i}} (\sqrt{\frac{c_1}{a_1}}, \dots, \sqrt{\frac{c_n}{a_n}})^T$. $\square$

Let us continue with (1.1). In formulating and attacking this problem, we are explicitly asking for an optimal solution of f over the set Ω . Practical reality, however, both from theoretical and computational viewpoints, dictates that we must in many circumstances be content with a second best.

Definition 1.1. A point $\mathbf{x}^* \in \Omega$ is called a local minimum point of f over Ω if there exists $\delta > 0$ such that $f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \forall \mathbf{x} \in \Omega$ with $\|\mathbf{x} - \mathbf{x}^*\| < \delta$.

An optimal solution of (1.1) is also called a global minimum point. Similarly, we can define local maximum point and global maximum point. Every local minimum or maximum point is called a local extremum.



- | | |
|-------------------|--------------------|
| • — local minimum | ⊙ — global minimum |
| × — local maximum | ⊗ — global maximum |

The following is a sufficient condition for local extrema of single variable functions.

Theorem 1.1. If $f^{(k)}(x^*) = 0$ for $k = 1, 2, \dots, n$, $f^{(n+1)}(x^*) \neq 0$, and $f^{(n+1)}(x)$ is continuous in a neighborhood of x^* , then x^* is a local extremum iff $n + 1$ is even. Furthermore, for even $n + 1$, if $f^{(n+1)}(x^*) > 0$ then x^* is a local minimum; if $f^{(n+1)}(x^*) < 0$ then x^* is a local maximum.

Proof. To justify the statements, let h be an arbitrary number with sufficiently small $|h|$ (keep in mind that h can be both positive and negative). According to Taylor's theorem, we have

$$f(x^* + h) - f(x^*) = \sum_{k=1}^n \frac{f^{(k)}(x^*)}{k!} h^k + \frac{f^{(n+1)}(x^* + \theta h)}{(n+1)!} h^{n+1}$$

for some $0 < \theta < 1$. It follows from the hypothesis that

$$\begin{aligned} f(x^* + h) - f(x^*) &= \frac{f^{(n+1)}(x^* + \theta h)}{(n+1)!} h^{n+1} \\ &= \frac{f^{(n+1)}(x^* + \theta h)}{(n+1)! f^{(n+1)}(x^*)} (f^{(n+1)}(x^*) h^{n+1}). \end{aligned} \quad (1.2)$$

By the continuity of $f^{(n+1)}(x)$, $f^{(n+1)}(x^* + \theta h)$ has the same sign as $f^{(n+1)}(x^*)$ for all h with sufficiently small $|h|$. Thus

$$\frac{f^{(n+1)}(x^* + \theta h)}{(n+1)! f^{(n+1)}(x^*)} > 0 \quad (1.3)$$

for all h with sufficiently small $|h|$.

Now we are ready to complete the proof.

Clearly, x^* is a local maximum iff $f(x^* + h) - f(x^*) \leq 0$ for all h with sufficiently small $|h|$ iff, by (1.2) and (1.3), $f^{(n+1)}(x^*) h^{n+1} \leq 0$ for all h with sufficiently small $|h|$ iff $n+1$ is even (why?) and $f^{(n+1)}(x^*) < 0$.

Similarly, x^* is a local minimum iff $f(x^* + h) - f(x^*) \geq 0$ for all h with sufficiently small $|h|$ iff, by (1.2) and (1.3), $f^{(n+1)}(x^*) h^{n+1} \geq 0$ for all h with sufficiently small $|h|$ iff $n+1$ is even and $f^{(n+1)}(x^*) > 0$. $\square$

To demonstrate the power of the above theorem, let us apply it to the following two problems.

EXAMPLE 1.5. Find all the local extrema of the function $f(x) = x^3 - 9x^2 + 27x - 27$.

SOLUTION. It is easy to see that

$$f'(x) = 3x^2 - 18x + 27,$$

$$f''(x) = 6x - 18,$$

$$f'''(x) = 6 \neq 0.$$

Setting $f'(x) = 0$ gives $x = 3$. Since $f'(3) = f''(3) = 0$ and $f'''(3) \neq 0$, from Theorem 1.1 we deduce that $x = 3$ is not a local extremum as $n+1 = 3$ is odd, and hence f has no local extremum as every such point satisfies $f'(x) = 0$. $\square$

EXAMPLE 1.6. Find all the local extrema of the function $f(x) = x^4 - 8x^3 + 24x^2 - 32x + 16$.

ANSWER: $x = 2$ is a local minimum, which is the unique local extremum. $\square$

Let us digress momentarily to survey some results in multivariable calculus. We shall use repeatedly the following conventions of notations and terminologies.

NOTATIONS:

- ∇f — The gradient of f , i.e., $\nabla f = (\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n})^T$.
- $\nabla^2 f$ — The Hessian Matrix of the 2^{nd} partial derivatives of f , i.e., $\nabla^2 f$ is the $n \times n$ matrix whose (i, j) entry is $\frac{\partial^2 f}{\partial x_i \partial x_j}$.
- $f \in C^k \iff f$ has continuous k^{th} partial derivatives.

Theorem 1.2. Let $f \in C^1$ and let $\mathbf{x}(\alpha) = \mathbf{x}^* + \alpha \mathbf{d}$ and $g(\alpha) = f(\mathbf{x}(\alpha))$. Then

$$g'(\alpha) = \nabla f(\mathbf{x}(\alpha))^T \mathbf{d}.$$

Proof. By definition

$$\begin{aligned} g(\alpha) &= f(x_1 + \alpha d_1, x_2 + \alpha d_2, \dots, x_n + \alpha d_n) \\ &\triangleq f(y_1, y_2, \dots, y_n), \end{aligned}$$

where $y_i = x_i + \alpha d_i$, $i = 1, 2, \dots, n$. It follows from the Chain Rule that

$$\begin{aligned} g'(\alpha) &= \frac{\partial f}{\partial y_1} \cdot \frac{dy_1}{d\alpha} + \frac{\partial f}{\partial y_2} \cdot \frac{dy_2}{d\alpha} + \dots + \frac{\partial f}{\partial y_n} \cdot \frac{dy_n}{d\alpha} \\ &= \frac{\partial f}{\partial y_1} \cdot d_1 + \frac{\partial f}{\partial y_2} \cdot d_2 + \dots + \frac{\partial f}{\partial y_n} \cdot d_n \\ &= \left(\frac{\partial f}{\partial y_1}, \dots, \frac{\partial f}{\partial y_n} \right) \begin{pmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{pmatrix} \\ &= \nabla f(\mathbf{x}(\alpha))^T \mathbf{d}. \end{aligned}$$

$\square$

Theorem 1.3. Let $f \in C^2$ and let $\mathbf{x}(\alpha) = \mathbf{x}^* + \alpha \mathbf{d}$ and $g(\alpha) = f(\mathbf{x}(\alpha))$. Then

$$g''(\alpha) = \mathbf{d}^T \nabla^2 f(\mathbf{x}(\alpha)) \mathbf{d}.$$

Proof. Recall the proof of the preceding theorem, we have

$$g'(\alpha) = \frac{\partial f}{\partial y_1} \cdot d_1 + \frac{\partial f}{\partial y_2} \cdot d_2 + \dots + \frac{\partial f}{\partial y_n} \cdot d_n.$$

From the Chain Rule, it can be seen that

$$\begin{aligned}
g''(\alpha) &= \sum_{i=1}^n \frac{\partial^2 f}{\partial y_1 \partial y_i} \frac{dy_i}{d\alpha} \cdot d_1 + \sum_{i=1}^n \frac{\partial^2 f}{\partial y_2 \partial y_i} \frac{dy_i}{d\alpha} \cdot d_2 + \cdots + \sum_{i=1}^n \frac{\partial^2 f}{\partial y_n \partial y_i} \frac{dy_i}{d\alpha} \cdot d_n \\
&= \sum_{i=1}^n \frac{\partial^2 f}{\partial y_1 \partial y_i} d_i d_1 + \sum_{i=1}^n \frac{\partial^2 f}{\partial y_2 \partial y_i} d_i d_2 + \cdots + \sum_{i=1}^n \frac{\partial^2 f}{\partial y_n \partial y_i} d_i d_n \\
&= \sum_{j=1}^n \sum_{i=1}^n \frac{\partial^2 f}{\partial y_j \partial y_i} d_j d_i = (d_1, \dots, d_n) \left[\frac{\partial^2 f}{\partial y_i \partial y_j} \right] \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix} \\
&= \mathbf{d}^T \nabla^2 f(\mathbf{x}(\alpha)) \mathbf{d}.
\end{aligned}$$

□

Theorem 1.4. (*Taylor's Theorem: Multidimensional Case*)

Let $f \in C^2$. Then

$$f(\mathbf{x}^* + \mathbf{d}) = f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^* + \theta \mathbf{d}) \mathbf{d}$$

for some $0 \leq \theta \leq 1$.

Proof. Let $g(\alpha) = f(\mathbf{x}^* + \alpha \mathbf{d})$. Then by Theorems 1.2 and 1.3, we have $g'(\alpha) = \nabla f(\mathbf{x}^* + \alpha \mathbf{d})^T \mathbf{d}$ and $g''(\alpha) = \mathbf{d}^T \nabla^2 f(\mathbf{x}^* + \alpha \mathbf{d}) \mathbf{d}$. Thus from Taylor's theorem, it follows that

$$\begin{aligned}
g(1) &= g(0) + g'(0)(1 - 0) + \frac{1}{2} g''(\theta)(1 - 0)^2 \\
&= f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^* + \theta \mathbf{d}) \mathbf{d}.
\end{aligned}$$

Or

$$f(\mathbf{x}^* + \mathbf{d}) = f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^* + \theta \mathbf{d}) \mathbf{d}$$

for some $0 \leq \theta \leq 1$.

□

Let us go back to the concept of local extrema. To derive necessary conditions satisfied by a local extremum, the basic idea is to consider the movement away from the point in some given direction. Along any given direction the objective function can be viewed as a function of a single variable, the parameter defining movement in this direction, and hence the ordinary calculus of a single variable is applicable.

Definition 1.2. Let $\mathbf{x} \in \Omega$. A vector $\mathbf{d}$ is called a feasible direction at $\mathbf{x}$ if there exists $\delta > 0$ such that $\mathbf{x} + \alpha \mathbf{d} \in \Omega$ for any $0 \leq \alpha \leq \delta$.

Theorem 1.5. (*First-Order Necessary Conditions*)

Let $f \in C^1$ be a function on $\Omega \subseteq \mathbb{R}^n$. If $\mathbf{x}^*$ is a local minimum point of f over Ω , then $\nabla f(\mathbf{x}^*)^T \mathbf{d} \geq 0$ for any feasible direction $\mathbf{d} \in \mathbb{R}^n$ at $\mathbf{x}^*$.

Proof. By definition, $\exists \delta > 0$ such that $\mathbf{x}^* + \alpha \mathbf{d} \in \Omega \quad \forall 0 \leq \alpha \leq \delta$. Set $\mathbf{x}(\alpha) = \mathbf{x}^* + \alpha \mathbf{d}$ and define $g(\alpha) = f(\mathbf{x}(\alpha))$ for $0 \leq \alpha \leq \delta$. Then $g(\alpha) \geq g(0)$ for all sufficiently small α as $\mathbf{x}^*$ is a local minimum point. Hence

$$g'(0) = \lim_{\alpha \rightarrow 0^+} \frac{g(\alpha) - g(0)}{\alpha} \geq 0.$$

The desired conclusion follows instantly from Theorem 1.2 as $g'(0) = \nabla f(\mathbf{x}(0))^T \mathbf{d} = \nabla f(\mathbf{x}^*)^T \mathbf{d}$. $\square$

Definition 1.3. A point $\mathbf{x}$ is called an interior point of $\Omega \subseteq \mathbb{R}^n$ if there exists $\delta > 0$, such that $\mathbf{y} \in \Omega$ for all $\mathbf{y}$ in $\mathbb{R}^n$ satisfying $\|\mathbf{x} - \mathbf{y}\| < \delta$.

REMARK. If $\mathbf{x}$ is an interior point of Ω , then any nonzero vector $\mathbf{d}$ in $\mathbb{R}^n$ is a feasible direction at $\mathbf{x}$.

Corollary 1.5.1. (*Unconstrained Case*)

Let $f \in C^1$ be a function on $\Omega \subseteq \mathbb{R}^n$. If $\mathbf{x}^*$ is a local minimum point of f over Ω and $\mathbf{x}^*$ is an interior point of Ω , then $\nabla f(\mathbf{x}^*) = 0$.

Proof. Since $\mathbf{x}^*$ is an interior point of Ω , any $\mathbf{d} \in \mathbb{R}^n$ is a feasible direction at $\mathbf{x}^*$. By Theorem 1.5, we have $\nabla f(\mathbf{x}^*)^T \mathbf{d} = 0$ (why ?), which implies $\nabla f(\mathbf{x}^*) = 0$. $\square$

REMARK. This condition leads to n equations in n variables, which can be used to determine a solution in many cases.

Theorem 1.6. (*Second-Order Necessary Conditions*)

Let $f \in C^2$ be a function on $\Omega \subseteq \mathbb{R}^n$ and let $\mathbf{x}^*$ be a local minimum point. Then for any feasible direction $\mathbf{d} \in \mathbb{R}^n$ at $\mathbf{x}^*$, we have

- (i) $\nabla f(\mathbf{x}^*)^T \mathbf{d} \geq 0$
- (ii) $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq 0$ whenever $\nabla f(\mathbf{x}^*)^T \mathbf{d} = 0$.

Proof. (i) follows from Theorem 1.5.

(ii) By hypothesis, $\exists \delta > 0$ such that $\mathbf{x}^* + \alpha \mathbf{d} \in \Omega \quad \forall 0 \leq \alpha \leq \delta$. Define $\mathbf{x}(\alpha) = \mathbf{x}^* + \alpha \mathbf{d}$ and $g(\alpha) = f(\mathbf{x}(\alpha))$ for any $0 \leq \alpha \leq \delta$. Then by Theorems 1.2 and 1.3, we have $g'(0) = \nabla f(\mathbf{x}^*)^T \mathbf{d} = 0$ and $g''(0) = \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d}$.

Since $\mathbf{x}^*$ is a local minimum point, $g(\alpha) \geq g(0)$ whenever α is sufficiently small. Hence

$$\begin{aligned} 0 &\leq \lim_{\alpha \rightarrow 0^+} \frac{g(\alpha) - g(0)}{\alpha^2} = \lim_{\alpha \rightarrow 0^+} \frac{g'(\alpha)}{2\alpha} = \lim_{\alpha \rightarrow 0^+} \frac{g'(\alpha) - g'(0)}{2(\alpha - 0)} \\ &= \frac{1}{2}g''(0). \end{aligned}$$

It follows that $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq 0$. □

Corollary 1.6.1. (*Second-Order Necessary Conditions— Unconstrained Case*)

Let $f \in C^2$ be a function on $\Omega \subseteq \mathbb{R}^n$ and let $\mathbf{x}^*$ be an interior local minimum point. Then

(i) $\nabla f(\mathbf{x}^*) = 0$

(ii) $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq 0 \quad \forall \mathbf{d} \in \mathbb{R}^n$, i.e., $\nabla^2 f(\mathbf{x}^*)$ is positive semi-definite.

EXAMPLE 1.7. Show that no interior point is an optimal solution to the following optimization problem

$$\begin{aligned} \text{Min} \quad & f(x_1, x_2) = x_1^3 - x_1^2 x_2 + 2x_2^2 \\ \text{s.t.} \quad & x_1 \geq 0 \text{ and } x_2 \geq 0. \end{aligned}$$

Proof. Setting $\frac{\partial f}{\partial x_1} = \frac{\partial f}{\partial x_2} = 0$ (recall Corollary 1.5.1) yields

$$\begin{cases} 3x_1^2 - 2x_1 x_2 = 0 \\ -x_1^2 + 4x_2 = 0. \end{cases}$$

We thus get two solutions $(0, 0)$ and $(6, 9)$. Since $(0, 0)$ is a boundary point of the feasible region, we only need to consider $(6, 9)$.

Clearly $\nabla^2 f = \begin{bmatrix} 6x_1 - 2x_2 & -2x_1 \\ -2x_1 & 4 \end{bmatrix}$, so $\nabla^2 f|_{(6,9)} = \begin{bmatrix} 18 & -12 \\ -12 & 4 \end{bmatrix}$, which is not positive semi-definite as its determinant is negative. It follows from Corollary 1.6.1 that $(6, 9)$ is not an optimal solution, completing the proof. □

QUESTION 1.3. Does f have any local minimum point in the feasible region? Justify your answer.

Definition 1.4. A point $\mathbf{x}^* \in \Omega$ is called a strict local minimum point of f over Ω if $\exists \delta > 0$ such that $f(\mathbf{x}^*) < f(\mathbf{x}) \quad \forall \mathbf{x} \in \Omega$ with $0 < \|\mathbf{x} - \mathbf{x}^*\| < \delta$.

Theorem 1.7. (*Second-Order Sufficient Conditions*)

Let $f \in C^2$ be a function on $\Omega \subseteq \mathbb{R}^n$ and let $\mathbf{x}^*$ be an interior point of Ω . If

(i) $\nabla f(\mathbf{x}^*) = 0$ and

(ii) $\nabla^2 f(\mathbf{x}^*)$ is positive definite,

then $\mathbf{x}^*$ is a strict local minimum point of f .

Proof. Our proof is based on the following fact: $\nabla^2 f(\mathbf{x}^* + \mathbf{d})$ is positive definite whenever $\|\mathbf{d}\|$ is sufficiently small.

Assume the fact, our proof goes as follows. By Theorem 1.4, we have

$$\begin{aligned} f(\mathbf{x}^* + \mathbf{d}) - f(\mathbf{x}^*) &= \nabla f(\mathbf{x}^*)^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^* + \theta \mathbf{d}) \mathbf{d}, \quad \text{for some } 0 < \theta < 1. \\ &= \frac{1}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^* + \theta \mathbf{d}) \mathbf{d}. \end{aligned}$$

In view of the above fact, $\nabla^2 f(\mathbf{x}^* + \theta \mathbf{d})$ is positive definite for all nonzero $\mathbf{d}$ with sufficiently small $\|\mathbf{d}\|$, and thus $\mathbf{d}^T \nabla^2 f(\mathbf{x}^* + \theta \mathbf{d}) \mathbf{d} > 0$, implying $f(\mathbf{x}^* + \mathbf{d}) - f(\mathbf{x}^*) > 0$. So $\mathbf{x}^*$ is a strict local minimum point of f .

Let us now turn to establishing the fact.

It suffices to prove that $\exists \delta > 0$ such that the determinant of each of the principal submatrices of $\nabla^2 f(\mathbf{x}^* + \mathbf{d})$ is greater than 0 if $\|\mathbf{d}\| < \delta$. Let H_k (resp. $H_k(\mathbf{d})$) denote the principal submatrix made up of the first k rows and columns of $\nabla^2 f(\mathbf{x}^*)$ (resp. $\nabla^2 f(\mathbf{x}^* + \mathbf{d})$), where $1 \leq k \leq n$. Then the determinant of $H_k(\mathbf{d})$ is a continuous function of $\mathbf{d}$. Thus

$$\lim_{\|\mathbf{d}\| \rightarrow 0} \det\{H_k(\mathbf{d})\} = \det\{H_k\} > 0.$$

Since there are n principal submatrices in total, $\exists \delta > 0$ such that $\det\{H_k(\mathbf{d})\} > 0$ whenever $\|\mathbf{d}\| < \delta$ for all $k = 1, 2, \dots, n$, completing the proof. $\square$

REMARK. If (ii) is replaced by “ $\nabla^2 f(\mathbf{x}^*)$ is positive semi-definite”, then the statement no longer holds. For instance, let $f(x) = x^3$. Then $f'(0) = f''(0) = 0$. However 0 is not a local minimum.

EXAMPLE 1.8. Find all the local minimum points of the function

$$f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 + x_1x_2 + x_2x_3 + x_3x_1 - 7x_1 - 8x_2 - 9x_3 + 101$$

SOLUTION. By Corollary 1.5.1, every local minimum point of $f(\mathbf{x})$ satisfies $\nabla f(\mathbf{x}) = 0$, or

$$\begin{cases} \frac{\partial f}{\partial x_1} = 2x_1 + x_2 + x_3 - 7 = 0 \\ \frac{\partial f}{\partial x_2} = x_1 + 2x_2 + x_3 - 8 = 0 \\ \frac{\partial f}{\partial x_3} = x_1 + x_2 + 2x_3 - 9 = 0. \end{cases}$$

Thus $\mathbf{x}^* = (1, 2, 3)$ is the unique solution. Note that $\nabla^2 f(\mathbf{x}^*) = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$ is positive definite as its eigenvalues are 1, 1 and 4 (all positive). By Theorem 1.7, $\mathbf{x}^*$ is the unique local minimum point. $\square$

In order to develop a theory directed toward characterizing global, rather than, local minimum points, it is necessary to introduce some sort of convexity assumptions. These results provide an interesting geometric interpretation of the second-order sufficiency result derived above.

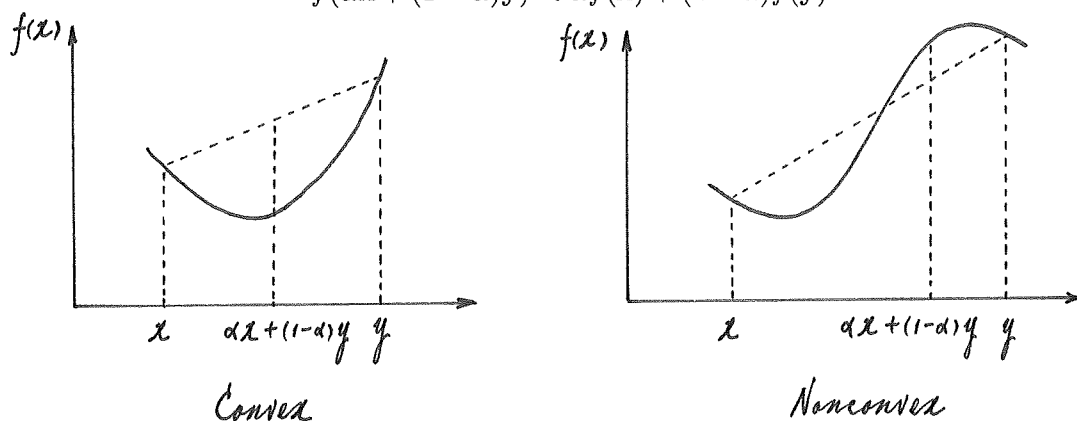
Definition 1.5. A set $\Omega \subseteq \mathbb{R}^n$ is called convex if for any two points $\mathbf{x}, \mathbf{y} \in \Omega$ and any $0 \leq \alpha \leq 1$, there holds $\alpha\mathbf{x} + (1 - \alpha)\mathbf{y} \in \Omega$.

Definition 1.6. A function f defined on a convex set Ω is said to be convex if, for every $\mathbf{x}, \mathbf{y} \in \Omega$ and every α , $0 \leq \alpha \leq 1$, there holds

$$f(\alpha\mathbf{x} + (1 - \alpha)\mathbf{y}) \leq \alpha f(\mathbf{x}) + (1 - \alpha)f(\mathbf{y}).$$

Function f is said to be strictly convex if, for every $\mathbf{x} \neq \mathbf{y}$ in Ω and every α , $0 < \alpha < 1$, there holds

$$f(\alpha\mathbf{x} + (1 - \alpha)\mathbf{y}) < \alpha f(\mathbf{x}) + (1 - \alpha)f(\mathbf{y}).$$



Geometrically, a function is convex if the line joining two points on its graph lies nowhere below the graph. Thinking of a function in two dimensions, it is convex if its graph is bowl shaped.

Theorem 1.8. For a convex function, we have the following statements:

- (i) A function f defined on a convex set Ω is convex if and only if for any integer $k \geq 2$, any $\mathbf{x}_1, \dots, \mathbf{x}_k \in \Omega$, and any nonnegative $\alpha_1, \dots, \alpha_k$ with $\sum_{i=1}^k \alpha_i = 1$, there holds

$$f\left(\sum_{i=1}^k \alpha_i \mathbf{x}_i\right) \leq \sum_{i=1}^k \alpha_i f(\mathbf{x}_i).$$

- (ii) Let f be a convex function defined on a convex set Ω . Then f is continuous at every interior point of Ω .

Proof. (i) exercise; (ii) not required. □

Theorem 1.9. Let f and g be two convex functions on the convex set Ω . Then the following statements hold:

- (i) $f + g$ is convex on Ω ;
- (ii) cf is convex for any constant $c > 0$;
- (iii) $\Gamma_c = \{\mathbf{x} : \mathbf{x} \in \Omega \text{ and } f(\mathbf{x}) \leq c\}$ is a convex set for any c .

Proof. (i) and (ii) are immediate.

(iii) Let $\mathbf{x}, \mathbf{y} \in \Gamma_c$. Then $f(\mathbf{x}) \leq c$, $f(\mathbf{y}) \leq c$, and $\mathbf{x} + (1 - \alpha)\mathbf{y} \in \Omega$ for any $0 \leq \alpha \leq 1$ since Ω is convex. By convexity, we have

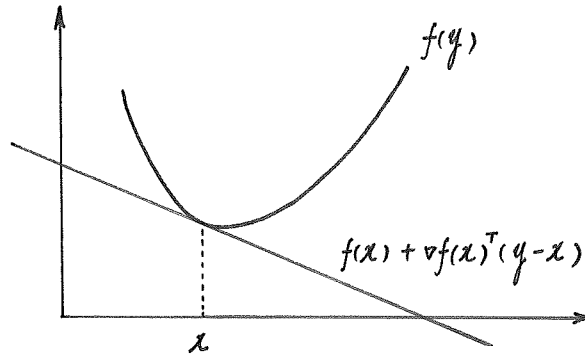
$$f(\alpha\mathbf{x} + (1 - \alpha)\mathbf{y}) \leq \alpha f(\mathbf{x}) + (1 - \alpha)f(\mathbf{y}) \leq c.$$

So $\alpha\mathbf{x} + (1 - \alpha)\mathbf{y} \in \Gamma_c$. □

Theorem 1.10. Let $f \in C^1$. Then f is convex over a convex set Ω iff

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x})$$

for all $\mathbf{x}, \mathbf{y} \in \Omega$.



Geometrically, the tangent plane going through a point is nowhere above the graph.

Proof. ($\implies$) If f is convex, then $\forall 0 < \alpha < 1$, we have

$$f(\alpha\mathbf{y} + (1 - \alpha)\mathbf{x}) \leq \alpha f(\mathbf{y}) + (1 - \alpha)f(\mathbf{x}).$$

Thus

$$\frac{f(\mathbf{x} + \alpha(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\alpha} \leq f(\mathbf{y}) - f(\mathbf{x}).$$

Letting $\alpha \rightarrow 0$ we obtain

$$\nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) - f(\mathbf{x}).$$

($\Leftarrow$) Assume $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in \Omega$, we aim to prove that for any $\mathbf{x}_1, \mathbf{x}_2 \in \Omega$ and $0 \leq \alpha \leq 1$, there holds

$$f(\alpha\mathbf{x}_1 + (1 - \alpha)\mathbf{x}_2) \leq \alpha f(\mathbf{x}_1) + (1 - \alpha)f(\mathbf{x}_2). \quad (1.4)$$

Setting $\mathbf{x} = \alpha\mathbf{x}_1 + (1 - \alpha)\mathbf{x}_2$ and $\mathbf{y} = \mathbf{x}_1$ or $\mathbf{y} = \mathbf{x}_2$ alternatively, we have

$$f(\mathbf{x}_1) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{x}_1 - \mathbf{x}) \quad (1.5)$$

$$f(\mathbf{x}_2) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{x}_2 - \mathbf{x}). \quad (1.6)$$

Multiplying (1.5) by α and (1.6) by $(1 - \alpha)$ and adding them, we have

$$\begin{aligned} \alpha f(\mathbf{x}_1) + (1 - \alpha)f(\mathbf{x}_2) &\geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T[\alpha\mathbf{x}_1 + (1 - \alpha)\mathbf{x}_2 - \mathbf{x}] \\ &= f(\mathbf{x}) \end{aligned}$$

as desired. □

REMARK. This theorem can be regarded as a sort of dual characterization of the original definition. The definition states that linear interpolation between any two points over-estimates the function. However, our theorem asserts that linear approximation based on the local derivative underestimates the function.

Theorem 1.11. *Let $f \in C^2$. Then f is a convex function over a convex set Ω containing an interior point iff the Hessian matrix $\nabla^2 f(\mathbf{x})$ of f is positive semi-definite throughout Ω .*

Proof. ($\Leftarrow$) For any $\mathbf{x}, \mathbf{y} \in \Omega$, by Taylor's Theorem we have

$$f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{1}{2}(\mathbf{y} - \mathbf{x})^T \nabla^2 f(\mathbf{x} + \alpha(\mathbf{y} - \mathbf{x}))(\mathbf{y} - \mathbf{x})$$

for some $0 < \alpha < 1$. Since $\nabla^2 f$ is positive semi-definite everywhere over Ω , $(\mathbf{y} - \mathbf{x})^T \nabla^2 f(\mathbf{x} + \alpha(\mathbf{y} - \mathbf{x}))(\mathbf{y} - \mathbf{x}) \geq 0$. Hence

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}).$$

It follows from Theorem 1.10 that f is convex.

($\Rightarrow$) Suppose the contrary: $\nabla^2 f$ is not positive semi-definite at some $\mathbf{x} \in \Omega$, that is, $\mathbf{d}^T \nabla^2 f(\mathbf{x}) \mathbf{d} < 0$ for some $\mathbf{d} \in \mathbb{R}^n$. The continuity of Hessian allows us to assume that $\mathbf{x}$ is an interior point of Ω (why?). This assumption together with $f \in C^2$ guarantee the existence of a ball $\delta(\mathbf{x})$ centered at $\mathbf{x}$ such that $\delta(\mathbf{x}) \subseteq \Omega$ and $\mathbf{d}^T \nabla^2 f(\mathbf{y}) \mathbf{d} < 0$ for

any $\mathbf{y} \in \delta(\mathbf{x})$. Now let us scale $\mathbf{d}$ by a positive number λ so that $\mathbf{x} + \frac{\alpha}{\lambda}\mathbf{d} \in \delta(\mathbf{x})$ for all $0 \leq \alpha \leq 1$. Then we have $\frac{\mathbf{d}^T}{\lambda} \nabla^2 f\left(\mathbf{x} + \frac{\alpha\mathbf{d}}{\lambda}\right) \frac{\mathbf{d}}{\lambda} < 0$ for all $0 \leq \alpha \leq 1$. Thus

$$\begin{aligned} f\left(\mathbf{x} + \frac{\mathbf{d}}{\lambda}\right) &= f(\mathbf{x}) + \nabla f(\mathbf{x})^T \frac{\mathbf{d}}{\lambda} + \frac{1}{2} \left(\frac{\mathbf{d}}{\lambda}\right)^T \nabla^2 f\left(\mathbf{x} + \alpha \frac{\mathbf{d}}{\lambda}\right) \frac{\mathbf{d}}{\lambda} \quad (\exists 0 \leq \alpha \leq 1) \\ &< f(\mathbf{x}) + \nabla f(\mathbf{x})^T \frac{\mathbf{d}}{\lambda}, \end{aligned}$$

contradicting Theorem 1.10. □

Corollary 1.11.1. *Let $f \in C^2$ be a single variable function defined over an interval Ω . If $f''(x) \geq 0$ for all $x \in \Omega$, then f is convex over Ω .*

Theorem 1.12. *Let $f \in C^2$ be a function defined over a convex set Ω . If the Hessian matrix $\nabla^2 f(\mathbf{x})$ of f is positive definite throughout Ω , then f is strictly convex over Ω .*

Proof. Exercise. □

Corollary 1.12.1. *Let $f \in C^2$ be a single variable function defined over an interval Ω . If $f''(x) > 0$ for all $x \in \Omega$, then f is strictly convex over Ω .*

Let us turn to some classical results concerning minimizations or maximizations of convex functions.

Theorem 1.13. *Let f be a convex function defined on a convex set Ω . Then*

- (i) *the set where f achieves the minimum is convex;*
- (ii) *any local minimum point of f is also a global minimum point.*

Geometrically, (i) implies that all the minimum points are located together.

Proof. (i) Let c be the minimum value of f over Ω . Then by Theorem 1.9 (iii), $\{\mathbf{x} : \mathbf{x} \in \Omega \text{ and } f(\mathbf{x}) \leq c\}$ is convex.

(ii) Suppose the contrary: some local minimum point $\mathbf{x}$ is not global minimum. Then there exists $\mathbf{y} \in \Omega$ such that $f(\mathbf{y}) < f(\mathbf{x})$. By local minimality of $\mathbf{x}$, there is a neighborhood $\delta(\mathbf{x})$ of $\mathbf{x}$ such that $f(\mathbf{z}) \geq f(\mathbf{x})$ for any $\mathbf{z} \in \delta(\mathbf{x})$. Now choose $0 < \alpha < 1$ so that $\mathbf{z} = \alpha\mathbf{x} + (1 - \alpha)\mathbf{y} \in \delta(\mathbf{x})$. Then $f(\mathbf{z}) = f(\alpha\mathbf{x} + (1 - \alpha)\mathbf{y}) \leq \alpha f(\mathbf{x}) + (1 - \alpha)f(\mathbf{y}) < f(\mathbf{x})$, a contradiction. □

The next theorem asserts that if f is continuously differentiable, then satisfaction of the first order necessary conditions are both necessary and sufficient for a point to be a global minimum point.

Theorem 1.14. Let $f \in C^1$ be convex on the convex set Ω . If there exists $\mathbf{x}^* \in \Omega$ such that

$$\nabla f(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) \geq 0 \quad \forall \mathbf{x} \in \Omega$$

then $\mathbf{x}^*$ is a global minimum of f over Ω .

Proof. By Theorem 1.10, for any $\mathbf{x} \in \Omega$, we have

$$f(\mathbf{x}) \geq f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) \geq f(\mathbf{x}^*).$$

□

REMARK. For convex functions, the first-order necessary conditions are both necessary and sufficient for a point to be a global minimum.

EXAMPLE 1.9. Solve the following problem

$$\text{Min } f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T A \mathbf{x} - \mathbf{b}^T \mathbf{x},$$

where A is an $n \times n$ positive definite matrix.

SOLUTION. Clearly $\nabla f(\mathbf{x}) = A\mathbf{x} - \mathbf{b}$ and $\nabla^2 f(\mathbf{x}) = A$. Since A is positive definite, by Theorem 1.11, $f(\mathbf{x})$ is a convex function. Notice that $\mathbf{x}^* = A^{-1}\mathbf{b}$ is the unique solution to $\nabla f(\mathbf{x}) = 0$. By Theorem 1.14 and Corollary 1.5.1, $\mathbf{x}^*$ is the unique optimal solution. □

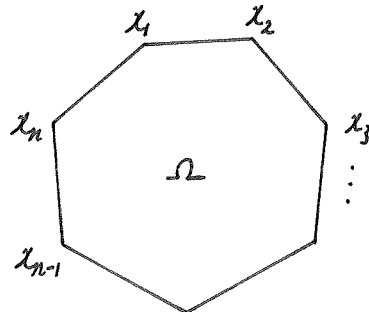
QUESTION 1.4. Let A be an $m \times n$ matrix of rank n and let $\mathbf{b} \in \mathbb{R}^m$. What is the solution to the following problem

$$\text{Min } \|A\mathbf{x} - \mathbf{b}\|?$$

Definition 1.7. A point $\mathbf{x}$ in a convex set Ω is said to be an extreme point of Ω if there exist no two points $\mathbf{x}_1$ and $\mathbf{x}_2$ in Ω such that $\mathbf{x} = \alpha\mathbf{x}_1 + (1 - \alpha)\mathbf{x}_2$ for some $0 < \alpha < 1$.

Theorem 1.15. Let f be a convex function defined on a bounded, closed convex set Ω . If f has a maximum value over Ω , then it is achieved at an extreme point of Ω .

Proof. Omitted. To facilitate better understanding of this theorem, let us take Ω as shown below.



Then every point $\mathbf{x}$ in Ω can be written as $\sum_{i=1}^n \alpha_i \mathbf{x}_i$, where $0 \leq \alpha_i \leq 1$ and $\sum_{i=1}^n \alpha_i = 1$.

By Theorem 1.8, we have $f(\mathbf{x}) \leq \sum \alpha_i f(\mathbf{x}_i) \leq \max_{1 \leq i \leq n} f(\mathbf{x}_i)$, so f attains maximum at an extreme point. $\square$

The remainder of this course is devoted to presentation and analysis of various algorithms designed to solve (1.1). The algorithms in this course usually go as follows: start at a point $\mathbf{x}_0 \in \Omega$. In general, suppose we have already had $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_k$ at hand. Then we calculate $\mathbf{x}_{k+1}$ according to certain principle so that

$$\lim_{k \rightarrow \infty} f(\mathbf{x}_k) = \text{the optimal value of (1.1)}.$$

Although these algorithms vary substantially in their motivation, application, and detailed analysis, ranging from the simple to highly complex, they have the common heritage of all being iterative descent algorithms.

- Iterative — the algorithm generates a series of points, each point being calculated on the basis of the points preceding it.
- Descent — as each new point is generated by the algorithm, the corresponding value of some function decreases.

An algorithm is said to be globally convergent if for any starting point, the algorithm is guaranteed to generate a sequence of points converging to a solution.

Definition 1.7. Let the sequence $\{\mathbf{x}_k\}$ converge to $\mathbf{x}^*$. The order of convergence of $\{\mathbf{x}_k\}$ is defined as the supremum of the nonnegative numbers p satisfying

$$0 \leq \overline{\lim}_{k \rightarrow \infty} \frac{\|\mathbf{x}_{k+1} - \mathbf{x}^*\|}{\|\mathbf{x}_k - \mathbf{x}^*\|^p} = \beta < \infty. \quad (1.7)$$

To ensure that the definition is applicable to any sequence, it is stated in terms of limit superior rather than just limit. The sequence is said to converge with order i if (1.7) holds for $p = i$. Usually, the sequence is said to have linear convergence if (1.7) holds for $p = 1$ and $\beta < 1$, and said to have superlinear convergence if (1.7) holds for $p > 1$, or for $p = 1$ and $\beta = 0$.

REMARK. If a sequence converges with order p , then it also converges with order p' for any $p' < p$ (why?). That is why the order of convergence is defined in terms of the supremum.

EXAMPLE 1.10. Find the order of convergence for each of the following sequences $\{x_k\}$.

- (i) $x_k = a^k$, $0 < a < 1$. (ii) $x_k = a^{2^k}$, $0 < a < 1$.
 (iii) $x_k = \frac{1}{k}$. (iv) $x_k = \frac{1}{k^k}$.

SOLUTION: (i) For any $p \geq 0$, we have

$$\overline{\lim}_{k \rightarrow \infty} \frac{x_{k+1}}{x_k^p} = \overline{\lim}_{k \rightarrow \infty} \frac{a^{k+1}}{a^{pk}} = \overline{\lim}_{k \rightarrow \infty} a^{(1-p)k+1}.$$

Hence $\overline{\lim}_{k \rightarrow \infty} \frac{x_{k+1}}{x_k^p} = a$ if $p = 1$ and ∞ if $p > 1$. By definition, the order of convergence is 1; $\{x_k\}$ converges linearly to zero.

(ii) Since $\overline{\lim}_{k \rightarrow \infty} \frac{x_{k+1}}{x_k^p} = 1$ if $p = 2$ and ∞ if $p > 2$, the order of convergence of $\{x_k\}$ is 2; $\{x_k\}$ converges to zero with order 2.

(iii) The order of convergence is 1; $\{x_k\}$ is not linearly convergent.

(iv) The order of convergence is 1; $\{x_k\}$ converges to zero superlinearly. □